

```
hive> CREATE TABLE adult_managed (  
>   age INT,  
>   workclass STRING,  
>   fnlwgt INT,  
>   education STRING,  
>   education_num INT,  
>   marital_status STRING,  
>   occupation STRING,  
>   relationship STRING,  
>   race STRING,  
>   sex STRING,  
>   capital_gain INT,  
>   capital_loss INT,  
>   hours_per_week INT,  
>   native_country STRING,  
>   income STRING  
> )  
> ROW FORMAT DELIMITED  
> FIELDS TERMINATED BY ','  
> STORED AS TEXTFILE;
```

ip-172-31-38-63 31% 5.17 GB / 7.79 GB 0.01 Mb/s 0.00 Mb/s 79 min hadoop /: 56% /emr: 1% /mnt: 3%

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```
hive> LOAD DATA INPATH '/user/hadoop/adult_data.txt'
> INTO TABLE adult_managed;
Loading data to table demo.adult_managed
OK
Time taken: 2.25 seconds
hive> █
```

 ip-172-31-38-63	 24%		 5.13 GB / 7.79 GB	 0.01 Mb/s	 0.01 Mb/s	 85 min	 hadoop	 /: 56%	/emr: 1%	/mnt:
---	---	---	---	---	---	--	--	--	----------	-------

hive> SELECT * FROM adult_external LIMIT 5;

39	State-gov	NULL	Bachelors	NULL	Never-married	Adm-clerical	Not-in-family	White	Male	NULL	NULL	NULL	United-States
<=50K	Self-emp-not-inc	NULL	Bachelors	NULL	Married-civ-spouse	Exec-managerial	Husband	White	Male	NULL	NULL	NULL	United-States
50	Private	<=50K	HS-grad	NULL	Divorced	Handlers-cleaners	Not-in-family	White	Male	NULL	NULL	NULL	United-States
38	Private	<=50K	11th	NULL	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	NULL	NULL	NULL	United-States
d-States	Private	<=50K	Bachelors	NULL	Married-civ-spouse	Prof-specialty	Wife	Black	Female	NULL	NULL	NULL	Cuba
53	Private	<=50K	Bachelors	NULL	Married-civ-spouse	Prof-specialty	Wife	Black	Female	NULL	NULL	NULL	Cuba
d-States	Private	<=50K	Bachelors	NULL	Married-civ-spouse	Prof-specialty	Wife	Black	Female	NULL	NULL	NULL	Cuba
28	Private	<=50K	Bachelors	NULL	Married-civ-spouse	Prof-specialty	Wife	Black	Female	NULL	NULL	NULL	Cuba

Time taken: 2.351 seconds, Fetched: 5 row(s)
hive>

```
hive> DROP TABLE adult_managed;
```

```
OK
```

```
Time taken: 3.012 seconds
```

```
hive> █
```

```
OK
Time taken: 0.115 seconds
hive> CREATE TABLE adult_managed (
>   age INT,
>   workclass STRING,
>   fnlwgt INT,
>   education STRING,
>   education_num INT,
>   marital_status STRING,
>   occupation STRING,
>   relationship STRING,
>   race STRING,
>   sex STRING,
>   capital_gain INT,
>   capital_loss INT,
>   hours_per_week INT,
>   native_country STRING,
>   income STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
```

```
OK
Time taken: 0.097 seconds
hive> █
```

 ip-172-31-38-63	 3%	 5.24 GB / 7.79 GB	 0.01 Mb/s	 0.00 Mb/s	 101 min	 hadoop	 /: 56%	 /emr: 1%	 /mnt: 3%
--	--	---	---	---	---	--	--	--	--

```
adult_data.txt
hive> LOAD DATA INPATH '/user/hadoop/adult_data.txt'
> INTO TABLE adult_managed;
Loading data to table demo.adult_managed
OK
Time taken: 0.602 seconds
hive> █
```

 ip-172-31-38-63	 1%		 5.25 GB / 7.79 GB	 0.01 Mb/s	 0.01 Mb/s	 103 min	 hadoop	
---	--	---	---	---	---	---	--	---

```
hive> SELECT AVG(age) AS average_age
> FROM adult_managed;
Query ID = hadoop_20260428144340_bbf70f5c-9555-4bfb-b5e2-bbfeab263ac4
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1777381560883_0006)
```

VERTICES		MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	1	1	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0

```
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 6.76 s
```

```
OK
38.58164675532078
Time taken: 17.403 seconds, Fetched: 1 row(s)
```

```
hive> SELECT workclass, COUNT(*) AS total_people
> FROM adult_managed
> WHERE workclass != ' ?'
> GROUP BY workclass
> ORDER BY total_people DESC;
```

Query ID = hadoop_20260428145552_05697f14-aa34-446b-adf2-fa7b65fbb076

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container		SUCCEEDED	1	1	0	0	0	0
Reducer 2	container		SUCCEEDED	1	1	0	0	0	0
Reducer 3	container		SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.48 s

OK

Private	22696
Self-emp-not-inc	2541
Local-gov	2093
State-gov	1298
Self-emp-inc	1116
Federal-gov	960
Without-pay	14
Never-worked	7

Time taken: 7.452 seconds, Fetched: 8 row(s)

Time taken: 7.432 seconds, Fetched: 0 row(s)

hive> SELECT

```
> MIN(age) AS minimum_age,
> MAX(age) AS maximum_age
> FROM adult_managed;
```

Query ID = hadoop_20260428145621_d2869095-614f-46cb-8fd2-70305b34e541

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 5.92 s								

OK

17 90

Time taken: 6.56 seconds, Fetched: 1 row(s)

```
hive> SELECT
>   ROUND(
>     SUM(CASE WHEN income = ' >50K' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
>     2
>   ) AS percentage_income_above_50k
> FROM adult_managed;
Query ID = hadoop_20260428145649_e4a266e2-92c4-42b1-ab11-75ba31afc667
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container		SUCCEEDED	1	1	0	0	0	0
Reducer 2	container		SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 6.82 s

```
OK
24.08
Time taken: 7.514 seconds, Fetched: 1 row(s)
```

```
hive> SELECT occupation, AVG(hours_per_week) AS avg_hours
> FROM adult_managed
> WHERE occupation != ' ?'
> GROUP BY occupation
> ORDER BY avg_hours DESC
> LIMIT 5;
```

Query ID = hadoop_20260428145713_f2f7b8b5-31f4-4d1e-899f-c8a9da68248a

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.91 s

OK

```
Adm-clerical    NULL
Transport-moving  NULL
Craft-repair    NULL
Exec-managerial  NULL
Farming-fishing  NULL
```

Time taken: 7.94 seconds, Fetched: 5 row(s)

```
hive> SELECT education, COUNT(*) AS total_people
> FROM adult_managed
> GROUP BY education
> ORDER BY total_people DESC
> LIMIT 1;
```

Query ID = hadoop_20260428150053_17570d14-9da8-489c-8367-13a1894090d0

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0	
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0	

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 5.95 s

OK

HS-grad 10501

Time taken: 6.56 seconds, Fetched: 1 row(s)

```
hive> SELECT COUNT(*) AS married_bachelors_count
> FROM adult_managed
> WHERE marital_status LIKE ' Married%'
> AND education = ' Bachelors';
```

Query ID = hadoop_20260428150121_e4517ae5-a10b-4018-bba4-0619b6b925c1

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0

VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 6.90 s

OK

2840

Time taken: 7.592 seconds, Fetched: 1 row(s)

hive> SELECT race, sex, AVG(age) AS average_age
> FROM adult_managed
> GROUP BY race, sex
> ORDER BY race, sex;
Query ID = hadoop_20260428150133_c330b4bf-eb29-40bd-8914-107a9be966e3
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 1.01 s								

OK
NULL NULL NULL
Amer-Indian-Eskimo Female 37.11764705882353
Amer-Indian-Eskimo Male 37.208333333333336
Asian-Pac-Islander Female 35.089595375722546
Asian-Pac-Islander Male 39.073593073593074
Black Female 37.854019292604505
Black Male 37.68260038240918
Other Female 31.678899082568808
Other Male 34.65432098765432
White Female 36.811617681092336
White Male 39.65249817461145
Time taken: 1.779 seconds, Fetched: 11 row(s)

```

hive> SELECT
>   occupation,
>   ROUND(
>     SUM(CASE WHEN income = ' >50K' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
>     2
>   ) AS high_income_percentage
> FROM adult_managed
> WHERE occupation != ' ?'
> GROUP BY occupation
> ORDER BY high_income_percentage DESC
> LIMIT 1;
Query ID = hadoop_20260428150150_868fa932-3254-47b7-a2b2-a08e7028af7c
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 7.34 s								

```

OK
Exec-managerial      48.40
Time taken: 8.188 seconds, Fetched: 1 row(s)

```

```
hive> SELECT
>   occupation,
>   total_people,
>   RANK() OVER (ORDER BY total_people DESC) AS occupation_rank
> FROM (
>   SELECT occupation, COUNT(*) AS total_people
>   FROM adult_managed
>   WHERE occupation != ' ?'
>   GROUP BY occupation
> ) t;
```

Query ID = hadoop_20260428150223_50bed5c1-07f9-4379-a5d1-6a3d44653c41
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED

Map 1	container		SUCCEEDED	1	1	0	0	0	0
Reducer 2	container		SUCCEEDED	1	1	0	0	0	0
Reducer 3	container		SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.34 s

OK

Prof-specialty	4140	1	
Craft-repair	4099	2	
Exec-managerial	4066	3	
Adm-clerical	3770	4	
Sales	3650	5	
Other-service	3295	6	
Machine-op-inspct	2002	7	
Transport-moving	1597	8	
Handlers-cleaners	1370	9	
Farming-fishing	994	10	
Tech-support	928	11	
Protective-serv	649	12	
Priv-house-serv	149	13	
Armed-Forces	9	14	

Time taken: 7.015 seconds, Fetched: 14 row(s)


```
hive> SELECT percentile(age, 0.90) AS age_90th_percentile
> FROM adult_managed;
Query ID = hadoop_20260428150253_5ee51934-2609-4a39-833c-8b7268a1b19a
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 5.89 s								

```
OK
58.0
Time taken: 6.497 seconds, Fetched: 1 row(s)
hive> SELECT COUNT(*) AS overworked_low_earners
> FROM adult_managed
> WHERE hours_per_week > 40
> AND income = ' <=50K';
```

```
Query ID = hadoop_20260428150312_9f507f21-c86c-431a-8d25-2da9e34ddffe
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 7.94 s								

```
OK
0
Time taken: 10.976 seconds, Fetched: 1 row(s)
```

```
hive> WITH avg_table AS (  
>   SELECT AVG(age) AS avg_age, 1 AS join_key  
>   FROM adult_managed  
> ),  
> adult_table AS (  
>   SELECT age, 1 AS join_key  
>   FROM adult_managed  
> )  
> SELECT COUNT(*) AS above_average_age_count  
> FROM adult_table a  
> JOIN avg_table b  
> ON a.join_key = b.join_key  
> WHERE a.age > b.avg_age;
```

Query ID = hadoop_20260428150602_e90c8b8a-1846-4c4c-b536-30be5382ef07

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Map 3	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 4	container	SUCCEEDED	1	1	0	0	0	0	0
VERTICES: 04/04 [=====>>] 100% ELAPSED TIME: 11.30 s									

OK

15053

Time taken: 12.137 seconds, Fetched: 1 row(s)

```
hive> SELECT sex, income, COUNT(*) AS total_people
> FROM adult_managed
> GROUP BY sex, income
> ORDER BY sex, income;
Query ID = hadoop_20260428150638_73a7f310-4917-4535-aecc-e8ce2337641f
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)
```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.45 s

OK

NULL	NULL	1
Female	<=50K	9592
Female	>50K	1179
Male	<=50K	15128
Male	>50K	6662

Time taken: 7.098 seconds, Fetched: 5 row(s)

Time taken: 7.036 seconds, Fetched: 3 row(s)

```
hive> WITH education_age AS (  
> SELECT education, AVG(age) AS average_age  
> FROM adult_managed  
> GROUP BY education  
> )  
> SELECT education, average_age  
> FROM education_age  
> ORDER BY average_age DESC  
> LIMIT 3;
```

Query ID = hadoop_20260428150702_75fdf10d-a1c4-4c68-acc1-11ed873ad989

Total jobs = 1

Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1777381560883_0007)

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0	0
Reducer 3	container	SUCCEEDED	1	1	0	0	0	0	0

VERTICES: 03/03 [=====>>] 100% ELAPSED TIME: 6.07 s

OK

7th-8th 48.44582043343653

Doctorate 47.70217917675545

1st-4th 46.142857142857146

Time taken: 6.66 seconds, Fetched: 3 row(s)